

A Hybrid Transmission Scheme for Cell-Free Massive MIMO Systems with Phase Offset

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Abstract—Cell-Free massive multiple-input multiple-output (MIMO), which provides high spectral efficiency (SE) through the coherent joint transmission, has drawn much academic research interest. However, the implementation of coherent joint transmission is restricted by the phase offset caused by hardware defect and asynchronous reception. In this paper, by combining the advantages of the high performance provided by coherent joint transmission and the phase offset robustness provided by non-coherent joint transmission, we propose a hybrid transmission scheme based on access point (AP) grouping, which ensures decent coherent joint transmission by minimizing the phase offset between collaborative APs within the group, and avoids large phase offset between groups by executing non-coherent joint transmission. Moreover, to facilitate simulation verification of the performance of the proposed scheme under various preprocessing methods, we derive a generalized version of SE expression of the hybrid transmission scheme using successive interference cancellation. Furthermore, simulations are provided to verify the effectiveness of our proposed scheme.

Index Terms—Cell-Free massive MIMO, phase offset, coherent joint transmission, non-coherent joint transmission, hybrid transmission scheme.

I. INTRODUCTION

Cell-Free massive multiple-input multiple-output (MIMO), also known as distributed MIMO, is an emerging technology where multiple access points (APs) are connected to a central processing unit (CPU) and jointly serve a number of user equipments (UEs) in the same time-frequency resources [1]. Cell-Free massive MIMO systems are envisioned to increase coverage and spectral efficiency (SE) compared to traditional cellular networks without cooperation. The superior SE performance can be achieved by adopting coherent joint transmission, where the symbols sent to the UE are coherently transmitted from multiple adjacent serving APs [2], [3].

Implementing high performance coherent joint transmission in Cell-Free massive MIMO potentially requires strict phase synchronization between desired signals received by UE from cooperative APs. However, it is difficult for distributed communication systems. Firstly, although the mismatch between uplink (UL) and downlink (DL) antenna arrays of each AP caused by hardware defect and hardware damage can be calibrated by traditional co-localization MIMO calibration techniques [4], there may be more complex non-reciprocal problem in Cell-Free massive MIMO. Specifically, different

reference antennas and radio frequency (RF) chain gains of each AP may result in inter AP phase offset [5], [6], which can cause UE to stack the desired signals from collaborative APs reversely. Secondly, due to the different propagation distances, it is impossible for all transmitted signals to reach multiple UEs simultaneously. The delay phase generated by asynchronous reception is inevitable in distributed multi-transmitter multi-receiver scenarios, which further leads to phase offset of desired signal [7], [8]. Therefore, the phase offset affects the performance of coherent joint transmission and restricts the promotion of Cell-Free massive MIMO.

To address the above issues, various over-the-air calibration methods have been studied, including inter AP calibration [9], [10] and UE-assisted calibration [11], [12]. Specifically, the authors in [9] formulated the non-reciprocity calibration problem using the total least squares (TLS) model and solved it using singular value decomposition. To avoid the high overhead of TLS, the authors in [10] utilized the advantages of centralized Argos calibration algorithm with low computational complexity and extended it to Cell-Free massive MIMO systems through spanning tree method. However, inter AP calibration can only compensate the phase offset caused by mismatched RF chains on the AP side but not other various phase offset, such as delay phase. Thus, the authors in [11] proposed a dot-division calibration method, in which calibration coefficients were calculated at AP side by comparing the DL channel state information (CSI) reported by UE and the UL channel estimated through sounding reference signal (SRS). Besides, the authors in [12] proposed a weighted least squares calibration method with DL CSI feedback from the UE. Although the UE-assisted calibration can effectively restore the performance of coherent joint transmission, their feedback cost of DL CSI is enormous and increases with the expansion of the network size, which is difficult for the Cell-Free massive MIMO to bear. Moreover, due to the presence of non-ideal factors such as noise and quantization error, it is difficult to avoid the impact of phase offset solely through calibration algorithms in Cell-Free massive MIMO. The authors in [13] found that non-coherent joint transmission is not affected by phase offset while unable to obtain coherent superposition gain. To the best of our knowledge, realizing a transmission scheme that can simultaneously guarantee high SE performance and affordable overhead has not been well studied yet in Cell-Free massive MIMO with phase offset.

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To fill this gap, this paper combines the advantages of coherent and non-coherent joint transmission schemes and proposes a hybrid transmission scheme based on AP grouping. Specifically, cooperative APs within a group perform coherent joint transmission, while APs in different groups conduct non-coherent joint transmission. To further enhance the advantages of this transmission scheme, we propose a UE assisted grouping algorithm based on DL equivalent channels, which can achieve the trade-off between performance and feedback cost compared to UE assisted calibration. In addition, we derive a generalized version of SE expression of the hybrid transmission scheme using successive interference cancellation (SIC) to facilitate simulation verification of the performance of the proposed scheme under various preprocessing methods. Moreover, simulations are provided to demonstrate that the proposed hybrid transmission is nearly twice and multiple times higher than coherent joint transmission in terms of average SE and 95%-likely SE in severe phase offset, respectively.

The remainder of the paper is organized as follows. Section II introduces the Cell-Free massive MIMO system model. Section III presents the DL hybrid transmission mechanism. Numerical results and analysis are given in section IV and conclusions are made in section V.

II. SYSTEM MODEL

In this paper, we study a Cell-Free massive MIMO system comprising of M APs and K UEs. Besides, one single antenna and N antennas are deployed for each UE and AP, respectively. Moreover, we use the time-division duplex protocol in massive MIMO system because the measurement overhead of channel is much less than other duplex schemes. Then the channel frequency response between m th AP and k th UE at each coherence block, for $m=1, 2, \dots, M$ and $k=1, 2, \dots, K$, is modeled as independent Rayleigh fading [1], i.e.,

$$\mathbf{h}_{mk} \sim \mathcal{CN}(\mathbf{0}, \beta_{mk} \mathbf{I}_N) \quad (1)$$

where β_{mk} describes the large-scale fading determined by the geometric path loss and shadow fading. In order to purely evaluate the impact of phase offset on the transmission scheme, we model the actual DL channel as follows [14], [15]:

$$\mathbf{g}_{mk} = \mathbf{h}_{mk}^T e^{j\phi_{mk}} \quad (2)$$

where the phase offset ϕ_{mk} is modelled as uniform distribution, which is independent identically distributed.

A. User-Centric Cell-Free Network

Ideally, all APs within the Cell-Free network may jointly serve UEs in the same time-frequency resource, which is called a fully-connected architecture and has high macro diversity gain. However, due to the requirement of high fronthaul capacity, the fully-connected architecture is not scalable in terms of practical deployment [16]. Thus, the user-centric architecture is widely adopted at present, where a UE may only be served by a subset of APs.

According to the large-scale information between the UE and the AP, CPU performs clustering, i.e., determine the connection matrix $\mathbf{C} = [\mathbf{C}_1, \dots, \mathbf{C}_M] \in \mathbb{R}^{K \times M}$, where $\mathbf{C}_m =$

$[c_{m1}, \dots, c_{mK}]^T$. When the relative large-scale value of a pair of AP-UE is smaller than the predefined clustering threshold η , the element $c_{mk}=1$, represents the connection between m th AP and k th UE, otherwise $c_{mk}=0$ represents no connection. The cluster of APs serving k th UE D_k can be defined as

$$D_k = \left\{ m : 10\log_{10} \left(\frac{\beta_k^{\max}}{\beta_{mk}} \right) \leq \eta, \forall m \right\} \quad (3)$$

where $\beta_k^{\max}$ is the maximum of large-scale value of k th UE.

B. Downlink Preprocessing

DL preprocessing is employed to solve the problem of interference between multiple UEs and reduce the processing complexity and power consumption of the UE side. Firstly, the m th AP calculates the precoding matrix $\mathbf{W}_m^x = [\mathbf{w}_{m1}^x, \dots, \mathbf{w}_{mK}^x]$ locally, where x represents conjugate beamforming (CB) or zero forcing (ZF) precoding, i.e.,

$$\text{CB} : \mathbf{W}_m^{\text{cb}} = \bar{\mathbf{H}}_m^* \quad (4)$$

$$\text{ZF} : \mathbf{W}_m^{\text{zf}} = \bar{\mathbf{H}}_m^* (\bar{\mathbf{H}}_m^T \bar{\mathbf{H}}_m^*)^{-1} \quad (5)$$

where $\bar{\mathbf{H}}_m = \mathbf{H}_m \cdot \text{diag}(\mathbf{C}_m)$ is the UL partial channel matrix obtained by m th AP through receiving SRS, $\mathbf{H}_m = [\mathbf{h}_{m1}, \dots, \mathbf{h}_{mK}] \in \mathbb{C}^{N \times K}$ and $\text{diag}(\mathbf{C}_m)$ is a diagonal matrix with $\mathbf{C}_m$ as the main diagonal elements. Then the AP conducts normalization and equal power allocation to obtain the DL preprocessing matrix $\bar{\mathbf{W}}_m^x = [\bar{\mathbf{w}}_{m1}^x, \dots, \bar{\mathbf{w}}_{mK}^x]$, where

$$\bar{\mathbf{w}}_{mk}^x = \frac{\mathbf{w}_{mk}^x}{\sqrt{\|\mathbf{C}_m\|^2 \mathbb{E}\{\|\mathbf{w}_{mk}^x\|^2\}}}, c_{mk} = 1 \quad (6)$$

and $\mathbb{E}\{\cdot\}$ represents mean operation.

III. DOWNLINK HYBRID TRANSMISSION MECHANISM

In this section, we first analyze the impact of phase offset on coherent and non-coherent joint transmission, based on which, we propose a hybrid scheme of coherent and non-coherent transmission schemes to achieve the trade-off between high performance and strong robustness. In addition, to further enhance the advantages of the hybrid transmission scheme, we propose a UE assisted grouping algorithm based on DL equivalent channel. Moreover, we derive a generalized version of SE expression of the hybrid transmission scheme using SIC with arbitrary decoding order.

A. DL SE of Hybrid Transmission

To demonstrate the impact of phase offset on the DL performance of coherent joint transmission, we plot average SE versus standard deviation of phase offset for coherent and non-coherent transmission in Fig.1, where non-coherent joint transmission acting as the benchmark. We observe that as the standard deviation of phase offset increases, the average SE of coherent joint transmission gradually decreases and even is lower than that of non-coherent joint transmission, indicating that phase offset may lead to negative coherent superposition gain in coherent joint transmission.

To alleviate the impact of phase offset on the system performance and harvest the potential superiority of coherent joint transmission, we propose a hybrid transmission scheme

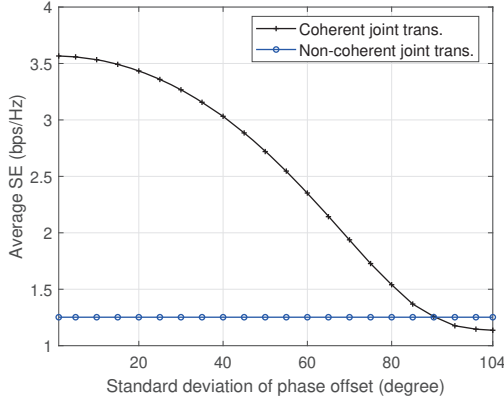


Fig. 1. Average SE versus standard deviation of phase offset for coherent and non-coherent transmission ($x = \text{cb}$, $M = 30$, $K = 8$, $N = 8$, and $\eta = 30\text{dB}$).

based on AP grouping. The core idea of the proposed scheme is to divide the service APs into G disjoint groups with UE as the center, where APs within a group send the same data symbol, i.e., perform coherent joint transmission, while APs in different groups send different data symbols, i.e., conduct non-coherent joint transmission. We assume that $q_{gk} \sim \mathcal{CN}(0, \rho_{gk})$ represents the data symbol sent by the APs in the g th group E_{gk} to the k th UE. The received signal of the k th UE is

$$\begin{aligned} r_k &= \sqrt{P_d} \sum_{k'=1}^K \sum_{g=1}^G \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk'}^x q_{gk'} + n_k \\ &= \sqrt{P_d} \sum_{g=1}^G \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x q_{gk} \\ &\quad + \sqrt{P_d} \sum_{k' \neq k}^K \sum_{g=1}^G \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk'}^x q_{gk'} + n_k \end{aligned} \quad (7)$$

where P_d is the maximum transmitting power of each AP and $n_k \sim \mathcal{CN}(0, \sigma_\omega^2)$ denotes the receiver noise at k th UE.

Since each UE has only a single antenna, the UE needs to employ SIC after receiving the desired signals from all G groups to detect the signals sent by the different AP groups. Specifically, the UE detects the signal transmitted by the g th group, and considers the signal sent from the $(g+1)$ th group to the G th group as uncorrelated noise. Based on this, the generalized version of SE expression of k th UE for the hybrid transmission scheme is shown at the top of the next page. The derivation is given in Appendix.

B. AP Grouping Based on DL Equivalent Channel

From the effective SE of hybrid transmission, we notice that the numerator contains the sum of squares of contributions from different groups, while the desired signals may still be stacked in reverse within different squares. To further explore the potential of hybrid transmission scheme and considering that there are partial phase offsets in distributed cooperative systems that must be handled with the help of UE, such as delay phase, etc., it is necessary to improve network service quality through UE assisted grouping. The specific steps are as follows:

Algorithm 1 AP grouping based on DL equivalent channel

- 1) The k th UE obtains a Cartesian coordinate form set of DL equivalent channel Ω_k by (11) and randomly selects G APs $\{u_1, \dots, u_G\}$ as the initial center point for each group. Set $j = 0$, $\Omega_k^j = \Omega_k - \{\mathbf{p}_{u_1k}, \dots, \mathbf{p}_{u_Gk}\}$. The g th group and its center point are $E_{gk}^j = \{u_g\}$ and $\mathbf{I}_{gk}^j = \mathbf{p}_{u_gk}$ in the j th iteration, respectively.
- 2) Other APs are assigned to the group with the smallest euclidean distance from the center point, i.e.,

$$g' = \underset{g=1, \dots, G}{\operatorname{argmin}} \left\| \mathbf{I}_{gk}^j - \mathbf{p}_{ik} \right\| \quad (9)$$

then $\Omega_k^j = \Omega_k^j - \{\mathbf{p}_{ik}\}$, $E_{g'k}^j = E_{g'k}^j + \{i\}$.

- 3) Recalculate the center point of each group $\mathbf{I}_{gk}^{j+1}$ by (12) when there are no elements in the set Ω_k^j .

- 4) Stop K-means algorithm, if $\mathbf{I}_{gk}^{j+1} = \mathbf{I}_{gk}^j, \forall g$. Otherwise, $j = j + 1$, $\Omega_k^j = \Omega_k$, $E_{gk}^j = \emptyset$ and go to step 2.

- 5) The AP closest to the center point in each group is selected as the master AP of the group. G master APs are $\{m_1, \dots, m_G\}$, $D_k = D_k - \{m_1, \dots, m_G\}$ and $R_{gk} = e_{m_gk}, \forall g$, where R_{gk} denotes the sum of complex signals of g th group;

- 6) AP $i \in D_k$ is assigned to the group that meet the criteria of maximizing the received power, i.e.,

$$g' = \underset{g=1, \dots, G}{\operatorname{argmax}} \left(|R_{gk} + e_{ik}|^2 + \sum_{q \neq g}^G |R_{qk}|^2 \right) \quad (10)$$

then $D_k = D_k - \{i\}$, $E_{g'k} = E_{g'k} + \{i\}$ and $R_{g'k} = R_{g'k} + e_{ik}$.

- 7) Stop if $D_k = \emptyset$. Otherwise, go to Step 6.

Firstly, the k th UE uses the DL beamformed reference signal (RS) to measure DL equivalent channel $e_{ik} = \mathbf{g}_{ik} \bar{\mathbf{w}}_{ik}^x, i \in D_k$ and converts the measured DL equivalent channel into the Cartesian coordinate form, i.e.,

$$\begin{aligned} \mathbf{p}_{ik} &= [p_{ik}^{(1)}, p_{ik}^{(2)}] \\ &= [\Re\{e_{ik}\}, \Im\{e_{ik}\}] \end{aligned} \quad (11)$$

where $\Re(\cdot)$ and $\Im(\cdot)$ denote the operations to obtain the real and imaginary parts, respectively.

Secondly, the UE randomly selects G APs as the initial center points for each group and other APs are assigned to the group with the smallest euclidean distance from the center point. Subsequently, the UE recalculates the center point of each group according to the following equation and reassigns the APs.

$$\mathbf{I}_{gk} = \left[\frac{\sum_{t \in E_{gk}} p_{tk}^{(1)}}{N_{gk}}, \frac{\sum_{t \in E_{gk}} p_{tk}^{(2)}}{N_{gk}} \right], g = 1, \dots, G \quad (12)$$

where $\mathbf{I}_{gk}$ and N_{gk} represent the center point and the number of AP in the group E_{gk} , respectively. When the center point of each group no longer changes, the above iterative steps stop, which is actually the well-known K-means algorithm.

$$SE_k^{hy} = \log_2 \left(1 + \frac{\sum_{g=1}^G \left| \mathbb{E} \left\{ \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right|^2 \rho_{gk}}{\sum_{k'=1}^K \sum_{g=1}^G \mathbb{E} \left\{ \left| \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right|^2 \right\} \rho_{gk'} - \sum_{g=1}^G \left| \mathbb{E} \left\{ \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right|^2 \rho_{gk} + \sigma_n^2} \right) \quad (8)$$

Finally, the UE selects the AP from each group that is closest to the center point of the output of K-means algorithm as the master AP. The remaining APs are grouped according to (10). The AP grouping can be summarized in algorithm 1.

C. Overall Process of Hybrid Transmission

By grouping AP based on DL equivalent channels, the flow of the hybrid transmission scheme is shown in Fig. 2.

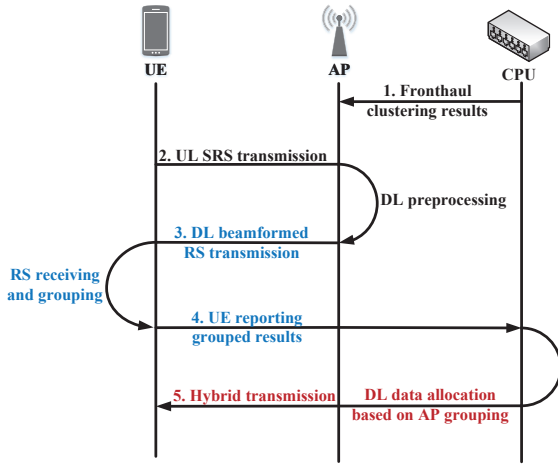


Fig. 2. Flow chart of the proposed hybrid transmission based on AP grouping.

Firstly, CPU forms a user-centric network containing multiple APs serving a UE. Secondly, each AP relies on the UL SRS transmitted by the UE to calculate the DL preprocessing matrix locally. Thirdly, each AP sends DL beamformed RS to UE it serves [17], then each UE receives DL beamformed RS and uses the algorithm 1 to divide the APs into G disjoint groups. In addition, each UE reports grouping results (e.g., feedback group index) to CPU. Finally, the AP side performs DL data hybrid transmission based on AP grouping, i.e., APs within a group send the same data symbols, while APs in different groups send different data symbols.

Note that phase offset caused by non-reciprocity generally depends on the hardware implementation, so it can be assumed to be constant over many propagation channel coherence intervals [5]. Besides, we assume that UE movement is slow, therefore the delay phase is also slowly changing. Thus, step 3–4 are typically performed infrequently in Fig. 2. In addition, compared to UE assisted calibration, the proposed scheme only needs to feed back 1-bit group index to each serving AP at $G=2$, which avoids the expensive cost of utilizing codebook feedback. Both of the above phenomena indicate that the cost of hybrid transmission is minor.

IV. NUMERICAL RESULTS AND ANALYSIS

We perform simulations to verify the effectiveness of our proposed method. The $M=30$ APs and $K=8$ UEs are randomly distributed within a square of size $800\text{m} \times 800\text{m}$. Besides, $N=8$ antennas are deployed for each AP. Three-slope propagation model is adopted in a simulation setup [1]. It is assumed that the bandwidth is 20 MHz and the carrier frequency is 1.9 GHz. In addition, the maximum transmitting power of each AP, clustering threshold, and the noise figure are set to 200 mW, 30 dB, and 9 dB, respectively. Additionally, the number of groups in the hybrid transmission scheme is 2. Considering the severe RF chains mismatch caused by low-cost hardware [5], [14] and unknown delay phase [8], we set the standard deviation of phase offset σ_p to $0 \sim 104^\circ$. Moreover, to demonstrate the advantages brought by the grouping algorithm proposed in the paper, we simulate the performance of the neighbor grouping, i.e., APs close to each other are assigned to different groups and the number of AP in each group is similar, which is proposed in [18] for space-time block codes to improve the coverage of system information broadcasting in Cell-Free massive MIMO. In addition, since a grouping result can be obtained by relying only on steps 1–4 of algorithm 1, we call it simplified algorithm 1 and simulate it to show the necessity of complete algorithm 1.

Fig. 3a shows average SE of different transmission schemes with different standard deviations of phase offset. We observe that the hybrid transmission based on neighbor grouping outperforms coherent joint transmission until $\sigma_p > 90^\circ$, while proposed schemes are superior to coherent joint transmission until $\sigma_p > 50^\circ$, where proposed schemes refer to the hybrid transmission based on simplified algorithm 1 and algorithm 1. On the one hand, this phenomenon is due to the fact that for neighbor grouping, the phase offset between APs participating in coherent joint transmission within the group may still be large and grouping can hardly improve coherent transmission performance, while proposed schemes are realized according to minimize the phase offset between APs within the group, which can achieve higher SE performance. On the other hand, APs between groups may have small phase offset for neighbor grouping, which loses the positive coherence gain, while proposed schemes are implemented based on ensuring maximum inter group phase offset, which can fully leverage the phase robustness of non-coherent joint transmission. Moreover, the gap of average SE between coherent joint transmission and proposed schemes gradually becomes larger as σ_p increases. Especially, the average SE of proposed schemes is increased by 98% with $\sigma_p=104^\circ$.

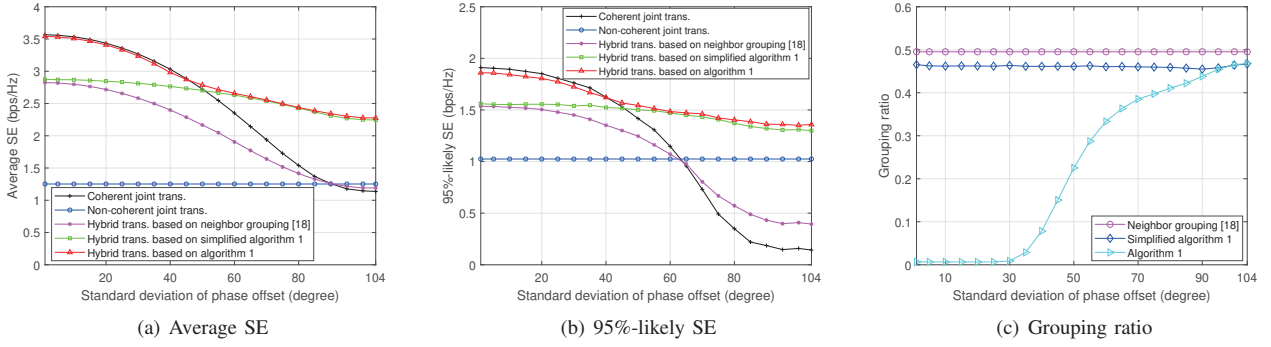


Fig. 3. Performance versus standard deviation of phase offset for different transmission under CB.

Fig. 3b shows 95%-likely SE of different transmission schemes with different standard deviations of phase offset, where the so-called 95%-likely SE refers to the performance that can be achieved by 95% of the UEs in the network. We notice that the hybrid transmission based on neighbor grouping is better than coherent joint transmission until $\sigma_p > 65^\circ$, and proposed schemes are superior to coherent joint transmission until $\sigma_p > 45^\circ$. Compared to Fig. 3a, it can be seen that the intersection points of coherent joint transmission and hybrid transmission schemes shift to the left. In addition, the 95%-likely SE of proposed schemes compared to coherent joint transmission is increased by 839% with $\sigma_p = 104^\circ$. Both of the above phenomena indicate that the proposed schemes perform better in improving edge performance.

Fig. 3c shows grouping ratio of different grouping schemes with different standard deviations of phase offset to explain the phenomenon in Figs. 3a and 3b where algorithm 1 outperforms the simplified algorithm 1 under small σ_p . The grouping ratio is defined as $\sum_{k=1}^K (\min\{N_{1k}, \dots, N_{Gk}\} / \sum_{g=1}^G N_{gk}) / K$. We observe that the grouping ratios are always around 0.49 and 0.46 for neighbor grouping and simplified algorithm 1, respectively, while the grouping ratio of algorithm 1 approaches 0 under smaller σ_p , indicating that the hybrid transmission based on algorithm 1 tends to degenerate into coherent joint transmission. The reason is that although simplified algorithm 1 can avoid generating negative coherent superposition gain by maximizing inter group phase offset under severe phase offset, which also leads to discard positive coherent superposition gain under small phase offset, resulting in poor performance. Based on this, the complete algorithm 1 avoids this situation by regrouping with the goal of maximizing the received power. Moreover, inspired by the above, algorithm 1 can be further improved. Specifically, we can set a grouping ratio threshold to achieve more flexible hybrid transmission, namely, when the grouping ratio is less than the set threshold, UE does not feedback the grouping result, which can save the feedback overhead. In other words, the hybrid transmission can achieve high performance as coherent joint transmission without the feedback overhead under smaller phase offset, while maintain high performance at the cost of feedback group index as phase offset increases, which achieves the trade-off between high SE performance and affordable overhead.

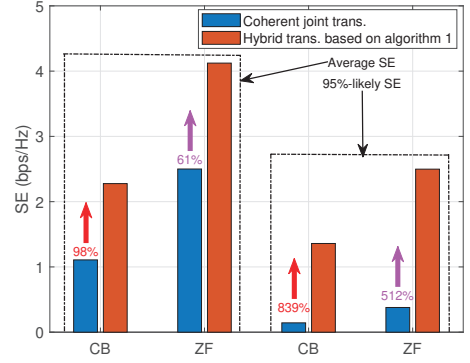


Fig. 4. Comparison of performance improvement under different precoding schemes with $\sigma_p = 104^\circ$.

Fig. 4 compares the performance improvement of hybrid transmission based on algorithm 1 under CB and ZF precoding with $\sigma_p = 104^\circ$. Since ZF precoding can eliminate the part of interference in user-centric Cell-Free network, it performs better than CB precoding. In addition, we observe that similar to CB precoding, our proposed scheme still has significant gain in terms of 95%-likely SE under ZF precoding, but its performance improvement decreases. This is because the ZF precoding sacrifices some degrees of freedom for interference cancellation which in turn leads to lower power of desired signal than CB precoding, while our proposed scheme aims to minimize the power of desired signal loss. Therefore, our proposed scheme is more necessary for CB precoding.

V. CONCLUSIONS

In this paper, to alleviate the impact of phase offset on the system performance, we have proposed a hybrid scheme of coherent and non-coherent transmission schemes. In addition, to further enhance the advantages of this transmission scheme, we have proposed a UE assisted grouping algorithm based on DL equivalent channels. Moreover, we have derived a generalized version of SE expression of the hybrid transmission scheme. As expected, the simulation results have verified the effectiveness of our proposed scheme.

APPENDIX

DERIVATION OF DL SE WITH HYBRID TRANSMISSION

The k th UE detects signal from g' th group by subtracting the first $g' - 1$ signals and only using the average channel $\mathbb{E}\{\sum_{m \in E_{g',k}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x\}$:

$$\sum_{g'=1}^G \log_2(1 + \text{SINR}_k^{(g')}) = \log_2 \left(\prod_{g'=1}^G \frac{\sum_{k'=1}^K \sum_{g=1}^G \mathbb{E} \left\{ \left| \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk'}^x \right|^2 \right\} \rho_{gk'} - \sum_{g=1}^{g'-1} \left| \mathbb{E} \left\{ \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right|^2 \rho_{gk} + \sigma_n^2}{\sum_{k'=1}^K \sum_{g=1}^G \mathbb{E} \left\{ \left| \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk'}^x \right|^2 \right\} \rho_{gk'} - \sum_{g=1}^{g'-1} \left| \mathbb{E} \left\{ \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right|^2 \rho_{gk} + \sigma_n^2} \right) \quad (16)$$

$$\text{SE}_k^{hy} = \log_2 \left(\frac{\sum_{k'=1}^K \sum_{g=1}^G \mathbb{E} \left\{ \left| \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk'}^x \right|^2 \right\} \rho_{gk'} + \sigma_n^2}{\sum_{k'=1}^K \sum_{g=1}^G \mathbb{E} \left\{ \left| \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk'}^x \right|^2 \right\} \rho_{gk'} - \sum_{g=1}^G \left| \mathbb{E} \left\{ \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right|^2 \rho_{gk} + \sigma_n^2} \right) \quad (17)$$

$$\begin{aligned} r_k^{(g')} &= r_k - \sqrt{P_d} \sum_{g=1}^{g'-1} \mathbb{E} \left\{ \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} q_{gk} \\ &= \sqrt{P_d} \mathbb{E} \left\{ \sum_{m \in E_{g'k}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} q_{g'k} \\ &\quad + \sqrt{P_d} \left(\sum_{m \in E_{g'k}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x - \mathbb{E} \left\{ \sum_{m \in E_{g'k}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right) q_{g'k} \\ &\quad + \sqrt{P_d} \sum_{g=1}^{g'-1} \left(\sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x - \mathbb{E} \left\{ \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right) q_{gk} \\ &\quad + \sqrt{P_d} \left[\sum_{g=g'+1}^G \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x q_{gk} + \sum_{k' \neq k} \sum_{g=1}^K \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk'}^x q_{gk'} \right] + n_k \end{aligned} \quad (13)$$

The first term in (13) is the desired signal part, and the rest are regarded as uncorrelated noise $\tilde{n}_{g'k}$ with power

$$\begin{aligned} \mathbb{E} \{ |\tilde{n}_{g'k}|^2 \} &= P_d \mathbb{E} \left\{ \left| \sum_{m \in E_{g'k}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x - \mathbb{E} \left\{ \sum_{m \in E_{g'k}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right|^2 \right\} \rho_{g'k} \\ &\quad + P_d \sum_{g=1}^{g'-1} \mathbb{E} \left\{ \left| \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x - \mathbb{E} \left\{ \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right|^2 \right\} \rho_{gk} \\ &\quad + P_d \sum_{g=g'+1}^G \mathbb{E} \left\{ \left| \sum_{m \in E_{gk}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right|^2 \right\} \rho_{gk} \\ &\quad + P_d \sum_{k' \neq k} \sum_{g=1}^K \mathbb{E} \left\{ \left| \sum_{m \in E_{gk'}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk'}^x \right|^2 \right\} \rho_{gk'} + \sigma_\omega^2 \end{aligned} \quad (14)$$

Note that these data symbols are mutually independent, and independent of all noise and channel coefficients. Thus, the DL SINR from g' th group is equal to

$$\text{SINR}_k^{(g')} = \frac{P_d \left| \mathbb{E} \left\{ \sum_{m \in E_{g'k}} \mathbf{g}_{mk} \bar{\mathbf{w}}_{mk}^x \right\} \right|^2 \rho_{g'k}}{\mathbb{E} \{ |\tilde{n}_{g'k}|^2 \}} \quad (15)$$

The total SE of k th UE is equal to (16), where $\sigma_n^2 \triangleq \sigma_\omega^2 / P_d$. After cancellations of terms that appear in both the numerator and denominator, we obtain (17). Finally, we obtain (8) after organizing the formulas.

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